

Siam Rahman Karip

 |  |  |  |  siam.rahman.karip@g.bracu.ac.bd

ABOUT

I am an independent researcher interested in large language models, spanning their evaluation, interpretability, and behavior across languages and settings. I remain open to a range of directions within this space and enjoy discovering new questions as I go.

EDUCATION

- **B.Sc in Computer Science and Engineering** Feb 2021 - Jan 2025
BRAC University Dhaka, Bangladesh
 - CGPA: 3.59/4.00 (Distinction)

WORK EXPERIENCE

- **Research Assistant (RA)** April 2025 – July 2026
United International University Dhaka, Bangladesh
Supervisor: Mr. Nahid Hossain, Assistant Professor and UG Program Coordinator
 - Conducting research in large language models and natural language processing, with a focus on understanding model behavior, evaluation, and reliability.
 - Conducting occasional classes, quizzes, and exam-script grading.
- **Private Tutoring** July 2022 – May 2025
 - Tutored university-level students in Python programming, covering core concepts and practical problem-solving.
 - Assisted A-Level students in Pure Mathematics and Statistics, strengthening analytical and quantitative skills.

RESEARCH INTERESTS

Large Language Models (LLMs), Natural Language Processing (NLP), Trustworthy Machine Learning

PUBLICATIONS

C=CONFERENCE, W=WORKSHOP

- [C.1] Nahid Hossain, **Siam Rahman Karip**, Md Faisal Kabir, Swakkhar Shatabda and Salekul Islam. 2026. **Systematic Failure Analysis of Open-Weight LLM Annotators for Classification Tasks: Taxonomy, Uncertainty Signals, and Deployment Boundaries**. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)* (Main Conference - To Appear), Budapest, Hungary. Association for Computational Linguistics.
- [W.1] **Siam Rahman Karip** and Nahid Hossain. 2026. **AlienAnnotators at PsyDefDetect: What Lies Between the Lines: Probing Lightweight Open-Source LLMs for Psychological Defense Mechanism Detection**. In *Proceedings of the of the BioNLP 2026 (Shared Tasks)*, pages 213–223, San Diego, California, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2026.bionlp-2.28

RESEARCH PROJECTS

- **Evaluation Protocol and Decoding Paradigm Effects on Semantic Reasoning in Diffusion vs. Autoregressive Language Models** In Progress
Target: ACL
 - Benchmarking diffusion and autoregressive language models across multiple evaluation protocols and decoding configurations on sentence-pair classification tasks requiring semantic reasoning.
 - Investigating how decoding paradigm and evaluation protocol choice jointly affect measured model performance and robustness, with a controlled comparison isolating architecture from training confounds.
- **Adapting Diffusion Language Models for Low-Resource Languages** In Progress
Target: ACL
 - Investigating the applicability of diffusion-based language modeling approaches to low-resource language settings, where training data scarcity poses distinct challenges.
 - Exploring architectural and training adaptations to improve data efficiency and downstream task performance relative to standard autoregressive baselines.
- **EEG and Eye-Tracking Feature Contextualization based Affective Computing through LightGBM enabled Stacked Ensembling of LLMs** Undergraduate Thesis
BRAC University
 - Introduced specific contextualized formatting for optimal tokenization.
 - Used LLMs (FLAN T5, GPT-2 and BERT) for classifying multimodal physiological numeric data.
 - Merged predictions from multiple LLMs using Majority Voting and Stacked Ensemble Learning Methods.

HONORS AND AWARDS

- **VC's List and Dean's List**

BRAC University

- Appeared on BRAC University's VC's list and Dean's list for last 3 semesters consecutively.

- **Academic Excellency Award**

Uttara North City Corporation

- Awarded for achieving a perfect GPA of 5.0 on all subjects in the Secondary School Board examination.

COURSES AND CERTIFICATIONS

- [] **Transformer Models with Pytorch** *Datacamp*
- [] **Large Language Models (LLMs) Concepts** *Datacamp*
- [] **AI Fundamentals** *Datacamp*
- [] **Intermediate Python** *Datacamp*

SKILLS

- **Programming Languages:** Python, C, Java, SQL, HTML, CSS, $\text{\LaTeX}$
- **Tools for Deep Learning:** PyTorch (CUDA), TensorFlow, Transformers (Hugging Face), BitsandBytes
- **LLM Tooling and Evaluation:** Hugging Face Datasets, Hugging Face Evaluate, Scikit-learn
- **Data Science and Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenCV
- **Version Control and Environment:** Git, GitHub

PROJECTS

- [] **Cross-Task Capability Testing of SLMs**
- [] **Sports Image Classification using a custom CNN model**
- [] **Email Spam Classification using ML models**
- [] **Credit Card Fraud Detection using ML models**

REFERENCES

1. **Nahid Hossain**

Assistant Professor and Program Coordinator
Department of Computer Science and Engineering
United International University
Email: nahid@cse.uiu.ac.bd
Relationship: Employer, Project Supervisor

2. **Swakkhar Shatabda**

Professor
Department of Computer Science and Engineering
BRAC University
Email: swakkhar.shatabda@bracu.ac.bd
Relationship: Project Co-supervisor

3. **Rafeed Rahman**

Senior Lecturer
Department of Computer Science and Engineering
BRAC University
Email: rafeed.rahman@bracu.ac.bd
Relationship: Thesis Supervisor